

Guideline Title: ST Elevation Myocardial Infarction (STEMI)

Guideline Number: GUID-3203-C010

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Revision dates: 4/30/24	Developed by: Air Care Team
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- I. **PURPOSE:** To improve the health and stability of the patient with an acute ST elevation myocardial infarction through timely recognition of specific electrocardiogram (ECG) abnormalities and early therapeutic intervention to reduce the overall extent of cardiac injury.
- II. **DEFINITIONS:**
 - a. STEMI - ST-Elevation Myocardial Infarction
- III. **DEPARTMENT GUIDELINE:**
 - a. Provide supplemental oxygen to maintain saturation > 93%.
 - b. Initiate cardiac monitoring, pulse oximetry, and vital signs every 5 minutes.
 - c. Expedited transport to the nearest percutaneous coronary intervention (PCI) capable hospital is of the utmost importance, do not delay transfer
 - d. Obtain/evaluate a 12 Lead ECG and interpret the findings. Assess for dysrhythmias and treat accordingly.
 - e. Ensure administration of **Aspirin 324 mg PO** (chewable) unless contraindicated by a true allergy.
 - f. Establish IV access (consider 2 sites).
 - g. If hypotension is present, and there is no evidence of pulmonary edema, consider 250 ml fluid bolus (0.9 % Normal Saline or Lactated Ringers). Repeat as necessary up to 500 ml.
 - h. If symptoms persist, consider administering **Nitroglycerine (NTG) 0.4 mg SL every 5 minutes x 3 doses** if hemodynamically stable. Use caution in inferior wall MI or right ventricular infarct in V4R.
 - i. If symptoms are relieved by SL NTG or if pain persists, consider initiating a **Nitroglycerin infusion at 10 mcg/min**, titrate until pain free, keeping MAP > 65 mmHg or SBP > 100 mmHg. Do not exceed 200 mcg/min.
 - ii. NTG is contraindicated if patient has received drugs for erectile dysfunction such as sildenafil (Viagra) or vardenafil (Levitra) within the past 24 hours and 72 hours for tadalafil (Cialis). (Consult medical direction).
 - i. For additional management for severe pain consider
 - i. **Morphine 0.1 mg/kg slow IV/IO** may repeat every 10 minutes PRN OR
 - ii. **Fentanyl 0.5-2 mcg/kg slow IV/IO** may repeat every 10 minutes PRN

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- iii. Monitor respiratory status and consider ETCO₂ monitoring.
- j. For anxiety management ensure adequate oxygenation then see *GUID3203-G006 Management for the Anxiety/Combative Patient*.
- k. For cocaine induced tachycardia, consider **Midazolam 0.05-0.1 mg/kg IV/IO**, may repeat every 3-5 minutes PRN, maximum dose 5 mg
- l. If hypotension persists, consider using vasopressors per *GUID3203-M011 Hypotension*.
- m. For significant tachyarrhythmias, refer to *GUID3203-C006 Narrow and Wide Complex Tachycardia*.
- n. If thrombolytic therapy has been initiated, continue infusion per referring physician's protocol. Monitor closely for bleeding.
- o. If the patient has not received low molecular weight heparin during the previous 12 hours and heparin therapy has not been initiated, consider administration of heparin after consulting with MD.
 - i. **Heparin bolus of 60 units/kg (max 5,000 units) followed by Heparin infusion at 18 units/kg (max 1,000 units per hour).**
- p. If glycoprotein IIB/IIIA (Integrilin) infusion has been initiated continue infusion per referring physician's protocol.
- q. Document administration of other antiplatelet agents which may include
 - i. Ticagrelor (Brilinta) 180 mg PO/NG
 - ii. Prasugrel (Effient) 60 mg PO/NG
 - iii. Clopidogrel 600 mg PO/NG in patients with high-risk bleeding or if others cannot be safely administered.
- r. If patient is not symptomatic, on low gtt doses of Nitroglycerine or Heparin, and flight time is < 10 minutes, consider discontinuing medications in favor of decreasing scene time if destination is a cardiac cath lab with interventional capabilities.

IV. DOCUMENTATION:

- a. EMS CHARTS
- b. Patient time of arrival to sending facility
- c. Time of STEMI on referring EKG
- d. Thrombolytic Therapy: Amount infused prior to arrival and amount infused during transport.

V. REFERENCES:

- a. American Heart Association, 2015.